

Digit readers will be familiar with the Crack The Code challenge and the time and effort required for the process of effectively cracking codes and ciphers. This process is called cryptanalysis, which is basically the science of analyzing information systems (commonly called codes and ciphers) in order to study the data hidden inside the systems. Cryptanalysis is used to beat cryptographic security systems, whether they are the pen-and-paper ones devised by you when you were kids, or the latest encrypted security systems anywhere in the world.

How does Cryptanalysis work?

There are four basic steps to solving any common cryptography puzzle. Although you don't need to stick to these steps strictly, they form a very useful guide for most beginners:

1. **Determine the language being used:** To decode the ciphertext of a coded message into plaintext, you should first have a general idea of what the plaintext is supposed to be like. So, it is important to first identify the language of the plaintext before attempting to look for it. That being said, there's not much to this step except your common sense. If you're decoding your friend's secrets, use the language he/she uses. If you're decoding a top-secret message from France, use French. Pretty simple.
2. **Identify the Encryption being used:** Certain Encryption systems are easily identifiable through the use of some telltale signs, while some others can be quite complex. However, irrespective of the complexity of the system used, once you have identified it, you have taken the first steps towards breaking it.
3. **Find the Key:** Most ciphers and codes use a certain 'key' that helps unlock them. Depending on the complexity of the Encryption system used, this process can be quite painstaking and laborious, but is often absolutely essential. Without a key, most cryptanalysis is simply reduced to brute-force guesswork and trial-and-error.
4. **Decode the Message:** Once you have the key and the ciphertext, you can decode the latter into plaintext. This is usually not the ultimate target of most cryptanalysts, who are simply interested in "cracking the code", unless of course the final message leads to buried treasure or something of the kind.

Most importantly, in the words of a Captain Parker Hitt from a U.S. Army cryptography textbook, "Success in dealing with unknown ciphers